

1 Data Generation

1.1 Pattern Matching

A *pattern* is a set.

1.1.1 Example

Let $a \circ b = [a_1, \dots, a_n, b_1, \dots, b_m]$, where $a = [a_1, \dots, a_n]$ and $b = [b_1, \dots, b_m]$.

$$\begin{aligned}\mathfrak{E} &:= \mathfrak{T} \cup \{\mathfrak{e}_1 \circ \mathfrak{o} \circ \mathfrak{e}_2 \mid \mathfrak{o} \in \{["+"], ["-"]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\ \mathfrak{T} &:= \mathfrak{F} \cup \{\mathfrak{t}_1 \circ \mathfrak{o} \circ \mathfrak{t}_2 \mid \mathfrak{o} \in \{["\times"], ["\div"]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\ \mathfrak{F} &:= \mathfrak{N} \cup \{["("] \circ \mathfrak{e} \circ [")"] \mid \mathfrak{e} \in \mathfrak{E}\} \\ \mathfrak{N} &:= N \cup N^* \\ N &:= \{ "0", \dots, "9" \}\end{aligned}$$

What is the difference between pattern matching and tokenwise parsing?

2 gen.jl and pattern.jl

The *map form* of a pattern $\mathfrak{P}$ is a function $\Pi : \mathfrak{P} \rightarrow L$, where $L \subseteq \mathcal{P}(\mathfrak{P})$ is a set of *labels*¹. The *inverse map form* of a pattern $\mathfrak{P}$ is a function $\Phi : L \rightarrow \mathcal{P}(\mathfrak{P})$, where $\Phi(l) = \{p \in \mathfrak{P} \mid \Pi(p) = l\}$ for $l \in L$.

Let $G := (\langle \mathcal{G} \rangle, *)$ be a group. Let $\Delta : \boxed{\Phi} \times L^2 \rightarrow \boxed{\Phi}$ – where $\boxed{\Phi}$ is the set of all inverse map forms of a pattern $\mathfrak{P} \subseteq G^{<\omega}$ and $L \subseteq G$ – be defined by $\forall l \in \text{dom}(\Phi) \setminus \{l_1 * l_2\} \quad (\Phi'(l) = \Phi(l))$ and $\Phi'(l_1 * l_2) = \Phi(l_1 * l_2) \cup \{[l_1, l_2]\}$ if $l_1 * l_2 \in \text{dom}(\Phi)$ and $\Phi'(l_1 * l_2) = \{[l_1, l_2]\}$ if not – where $\Phi' = \Delta(\Phi, [l_1, l_2])$ and $l_1, l_2 \in \text{dom}(\Phi)$.

Consider the function $\mathcal{R} : \boxed{\Phi} \times \mathbb{N} \rightarrow \boxed{\Phi}$ defined by

$$\mathcal{R}(\Phi, n) = \begin{cases} \Phi & : n = 0 \\ \mathcal{R} \left(\Delta \left(\Phi, \bigoplus_{i=1}^2 \text{Rand}(\text{dom}(\Phi)) \right) \right), n - 1 & : \text{otherwise} \end{cases} .$$

Now, let $\Phi_n = \mathcal{R}(\Phi, n)$, where $\Phi : \mathcal{G} \rightarrow \{\emptyset\}$ and $n \in \mathbb{N}$. Then Φ_n is a partial or full representation of the group G , where each element in $\text{dom}(\Phi_n)$ maps to the set of ordered pairs of elements in G to which it is mapped by $*$.

For each $l \in \text{dom}(\Phi_n)$, then, a set of vectors $\mathbf{s} \in \mathcal{G}^{<\omega}$ can be inferred with $\kappa : \boxed{\Phi} \times G \rightarrow \mathcal{G}^{<\omega}$!!!!!!!!! should $\mathcal{G}$ be included in domain too? probably, yes!!!!!!!!!, defined by

$$\kappa(\Phi, g) = \begin{cases} g & : g \in \mathcal{G} \\ \bigcup_{k \in \Phi(g)} \left(\bigoplus_{j \in k} \kappa(\Phi, j) \right) & : \text{otherwise} \end{cases}$$

!!!!!!!!!!!!THE DEFINITION IS WRONG, NOW THE VECTORS WILL INCLUDE SETS OF POSSIBLE VECTORS!!!!!!!!!!!!

¹Labes – or, *concepts* – are useful abstractions of patterns, which will be used later to optimize the complexity of solution-finding systems. The subsets $\mathfrak{T}, \mathfrak{F}, \mathfrak{N}$ and N in example 1.1.1 are examples of labes.